

AION 4-20 mA Adapter - Configuration Parameters

AION Resource Groups

Module	Group	Group ID	Description
AION	Timing	0x00	Measurement cadence and keep-alive uplink interval
AION	Configuration	0x01	Per-channel scaling, alarm, delta, and device/config id settings
AION	Telemetry	0x02	Read-only values published after a measurement
AION	Trigger	0x01	Manual measurement trigger executable

Kconfig Options

Kconfig symbol	Type	Default	Description
CONFIG_SENTOS_MODULE_AION	bool	n	Enables the Aion module

Timing - Group 0x00

Resource	ID	Description	Key	Min	Max	Default	Unit	Module key
MEASUREMENT PERIOD	0x00	Regular measurement period	period	1	660	1	min	XXX1
REGULAR SEND INTERVAL	0x01	Keep-alive uplink interval in measurements	every	1	24	1	measurements	XXX1

Configuration - Group 0x01

Resource	ID	Description	Key	Min	Max	Default	Unit	Module key
CHANNEL 1 UPPER BOUND	0x00	Channel 1 engineering value at 20 mA	upper1	-100000.0	100000.0	10.0	engineering value	XXX1
CHANNEL 2 UPPER BOUND	0x01	Channel 2 engineering value at 20 mA	upper2	-100000.0	100000.0	10.0	engineering value	XXX1
CHANNEL 1 ALARM THRESHOLD	0x02	Channel 1 alarm threshold; 0 disables alarm evaluation	thr1	-100000.0	100000.0	0.0	engineering value	XXX1
CHANNEL 2 ALARM THRESHOLD	0x03	Channel 2 alarm threshold; 0 disables alarm evaluation	thr2	-100000.0	100000.0	0.0	engineering value	XXX1
CHANNEL 1 ALARM HYSTERESIS	0x04	Channel 1 alarm hysteresis	hyst1	0.0	100000.0	0.0	engineering value	XXX1

Resource	ID	Description	Key	Min	Max	Default	Unit	Module key
CHANNEL 2 ALARM HYSTERESIS	0x05	Channel 2 alarm hysteresis	hyst2	0.0	100000.0	0.0	engineering value	XXX1
DEVICE / CONFIG ID	0x06	User-assigned device/config id, mirrored into telemetry	id	0	255	0	raw	XXX1
CHANNEL 1 LOWER BOUND	0x07	Channel 1 engineering value at 4 mA	lower1	-100000.0	100000.0	0.0	engineering value	XXX1
CHANNEL 2 LOWER BOUND	0x08	Channel 2 engineering value at 4 mA	lower2	-100000.0	100000.0	0.0	engineering value	XXX1
CHANNEL 1 DELTA	0x09	Channel 1 minimum scaled change since last uplink that forces a telemetry send; 0 disables delta sending	delta1	0.0	100000.0	0.0	engineering value	XXX1
CHANNEL 2 DELTA	0x0A	Channel 2 minimum scaled change since last uplink that forces a telemetry send; 0 disables delta sending	delta2	0.0	100000.0	0.0	engineering value	XXX1

Scaling 4-20 mA -> Engineering Value

Input current	Meaning
4 mA	Maps to lowerN
20 mA	Maps to upperN
Formula	$\text{scaled} = \text{lower} + ((\text{current_mA} - 4.0) / 16.0) * (\text{upper} - \text{lower})$

Alarm Behavior

Setting	Behavior
thrN = 0	Alarm evaluation is disabled for the channel
thrN != 0	Alarm trips when scaled value is greater than or equal to thrN
hystN	A latched alarm clears only after scaled value falls below thrN - hystN

Delta Behavior

Setting	Behavior
deltaN = 0	Delta-triggered sending is disabled for the channel
deltaN > 0	Telemetry is forced when absolute scaled change since last sent value is greater than or equal to deltaN

Trigger - Group 0x01

Resource	ID	Description	Key	Default	Input	Module key
TRIGGER MEASUREMENT	0x01	Triggers an irregular measurement. 0x0001 enables uplink sending; any other value performs a local-only user-triggered measurement.	measure	n/a	2-byte big-endian command value	XXX1